CHAPTER 18

Sequence, Series & Progressions

This is one of the chapters which contributes really very logical problems which can be solved easily by some innovative methods rather than conventional methods (like applying formulae etc.) Basically CAT is an exam where logic prevails over pure mathematics.

Therefore the students who use their ingenuity to solve these problems score better than those who rely on the formula based approach. In the starting of the chapter I have given the proper mathematical methods, then later on I have gradually moved towards novel techniques for faster and smarter calculation. Remember that almost 1–2 questions are necessarily asked in CAT and this chapter is also important for rest of the entrance tests eg. XAT, FMS, IIFT, CET, MAT etc. Once again I would like to advise you that maximum problems must be attempted with the help of given choices. Besides most of the problems can be solved by substituting the values of variables *e.g.,* n, x, etc. for nth term or sum of the series/progression. The above method is so smart that it reduces the time of problem solving upto 80% for some complex and typical problems.

Chapter Checklist

- Sequence
- Series
- Progressions
- Arithmetic Progression (AP)
- Geometric Progression (GP)
- Harmonic Progression
- Arithmetico-Geometric Sequence
- **CAT Test**

18.1 Sequence

A set of numbers occuring in a definite order or by a rule is called a sequence. Thus any set of numbers $a_1, a_2, a_3, \ldots, a_n$ such that to each positive integer n, there corresponds a number a_n, is sequence. The numbers $a_1, a_2, a_3, \ldots, a_n$ are called elements (or members) of the sequence.

e.g., $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{n}$ is a sequence, since the number occur in a definite order. A sequence can be finite or infinite according to the number of elements in it. If there are finite number of elements, then it is a finite sequence and if there are infinite number of elements in it, then it is an infinite sequence and is denoted by $\{a_n\}$.

18.2 Series

Let $u_1, u_2, u_3, \ldots, u_n$ be a given sequence of n numbers, where n is a specified positive integer. Then their sum *i.e.* $u_1 + u_2 + u_3 + \ldots + u_n$ is called a finite series and the numbers $u_1, u_2, u_3, \ldots, u_n$ are called terms of the series.

If $S_n = u_1 + u_2 + u_3 + \ldots + u_n$, then S_n is called the sum to n terms (or the sum of the first n terms) of

the series and is denoted by $\sum_{r=1}^{n} u_r$ or Σu_r. *e.g.*, $2+4+6+\ldots+30$ is a finite series and can be denoted as $\sum_{n=1}^{15}(2n)$ ($\Sigma \rightarrow$ Sigma, summation notation)

18.3 Progressions

Sequences of special types are called progressions. The sequences $a_1, a_2, \ldots, a_n \ldots$ or $\{a_n\}$ is known when its nth term a_n is given. **For example :**

(i) If $a_n = 2n$, then the sequence is 2, 4, 6, 8, …

(ii) If $a_n = 2n-1$, then the sequence is 1, 3, 5, 7, ….

But to define a sequence we may not always have a formula for the nth term. For example, the sequence of the prime numbers is 2, 3, 5, 7, 11, 13, … for which a formula for the nth term may not be given. In this chapter we study basically Arithmetic Progression (A.P.) and Geometric Progression (G.P.) and get a overview of Harmonic Progression (H.P.) also.

18.4 Arithmetic Progression (AP)

A sequence $a_1, a_2, a_3, \ldots, a_n$ is said to be in arithmetic progression, when $a_2 - a_1 = a_3 - a_2 = \ldots$ *i.e.,* when the quantities increase or decrease continuously by a common quantity and this constant (or common) quantity is called the common difference of the AP *e.g.,*

Progressions	Common Difference
1, 3, 5, 7, 9, …	$\rightarrow 2$
12, 9, 6, 3, 0, –3, – 6, …	$\rightarrow -3$
7, 12, 17, 22, …	$\rightarrow 5$
– 6, – 4, – 2, 0, 2, 4, 6, …	$\rightarrow 2$

General Term (or *n*th term) *of an AP*

Let us look at the following : 1, 4, 7, 10, 13, …

Each next term in the above sequence is being increased by 3 (which is the common difference). So you can see that from the second term onwards 3 is being added to every next term *i.e.,*

$1 = 1$
$4 = 1 + 3$
$7 = 1 + 3 + 3$
$10 = 1 + 3 + 3 + 3$
$13 = 1 + 3 + 3 + 3 + 3$ etc.

Let if a be the first term of the progression and d be the common difference, then the

First term $\longrightarrow a$
Second term $\longrightarrow a + d$
Third term $\longrightarrow a + 2d$
Fourth term $\longrightarrow a + 3d$
Fifth term $\longrightarrow a + 4d$

And nth term $\longrightarrow a + (n-1)d$

The general term or nth term is denoted by T_n.
Hence, $T_n = a + (n-1)d$.
Also $l = a + (n-1)d$ where l is the last term of AP

Exp. 1) Find the 11th term of $-7, -2, 3, 8, 13, \ldots$.

Solution $T_{11} = -7 + (11-1)5 = 43$ ($\because$ $a = -7$, $d = 5$ and $n = 11$)

Sum of First n terms of an AP Let us look at the following :
1, 4, 7, 10, 13, …
Let the sum of 'n' terms is denoted by S_n, then

$S_1 = 1$
$S_2 = 1 + 4 = 5$, $S_3 = 1 + 4 + 7 = 12$
$S_4 = 1 + 4 + 7 + 10 = 22$
$S_5 = 1 + 4 + 7 + 10 + 13 = 35$

$\therefore$ $S_n = T_1 + T_2 + T_3 + \ldots + T_n$
$= (a) + (a + d) + (a + 2d) + \ldots + (a + (n-1)d)$
$= na + \dfrac{n(n-1)d}{2} \Rightarrow S_n = \dfrac{n}{2}[2a + (n-1)d]$

Alternatively $S_n = \left(\dfrac{a + l}{2}\right)n$,

($l \rightarrow$ last term of the progression)

NOTE In my own experience the alternative formula of S_n is better for problem solving since in this formula we have to just take the product of average of first and last term with number of terms. Also for our convenience to use this formula we can consider any term as last term.

Exp. 2) Find the sum of 11 terms of $-7, -2, 3, 8, \ldots$

Solution $a = -7$

$l = -7 + (10)\cdot 5 = 43 \therefore$ $S_{11} = \dfrac{-7 + 43}{2} \times 11$

$S_{11} = 198$

NOTE Personally I hardly use the formula $S_n = \dfrac{n}{2}[2a + (n-1)d]$

Since, I prefer $S_n = \left(\dfrac{a + l}{2}\right)n$.

Sum of First n Terms of an A.P.

Let us look at the following :
1, 4, 7, 10, 13, …

Let the sum of 'n' terms is denoted by S_n, then

$S_1 = 1$
$S_2 = 1 + 4 = 5$
$S_3 = 1 + 4 + 7 = 12$
$S_4 = 1 + 4 + 7 + 10 = 22$
$S_5 = 1 + 4 + 7 + 10 + 13 = 35$

$\therefore$ $S_n = T_1 + T_2 + T_3 + \ldots + T_n$
$= (a) + (a + d) + (a + 2d) + \ldots + (a + (n-1)d)$
$= na + \dfrac{n(n-1)d}{2}$

$S_n = \dfrac{n}{2}[2a + (n-1)d]$

Arithmetic Mean (AM)

When three quantities are in AP, the middle one is called arithmetic mean (A.M.) of the other two.

Let a, m, b are in AP, then A.M. of a and b *i.e.*, $m = \frac{a+b}{2}$

$\therefore$ $\text{A.M.} = \frac{a+b}{2}$

In general, if $a, m_1, m_2, m_3, \ldots, m_n, b$ are in AP then the intermediate numbers $m_1, m_2, m_3, \ldots, m_n$ are called the n arithmetic means between a and b also the AM of $a_1, a_2, a_3, \ldots, a_n$, which are in AP is

$$\text{AM} = \frac{(a_1 + a_2 + a_3 + \ldots + a_n)}{n}$$

How to Insert *n* Arithmetic Means Between *a* and *b* If a and b are the first and last term respectively between which we have to insert n arithmetic means $m_1, m_2, m_3, \ldots, m_n$, then common difference $d = \frac{b-a}{n+1}$. Thus, $m_1 = a + \left(\frac{b-a}{n+1}\right)$

$\Rightarrow$ $m_2 = a + 2\left(\frac{b-a}{n+1}\right) \Rightarrow m_3 = a + 3\left(\frac{b-a}{n+1}\right)$

… … … … …

… … … … …

$m_n = a + n\left(\frac{b-a}{n+1}\right)$

Exp. 3) Insert 3 arithmetic means between 5 and 21.

Solution $d = \frac{21-5}{3+1} = 4$

$\therefore$ First A.M. $= a + d = 5 + 4 = 9$

Second A.M. $= a + 2d = 5 + 2(4) = 13$

Third A.M. $= a + 3d = 5 + 3(4) = 17$

Hence the new sequence is 5, 9, 13, 17, 21, where all the terms are in AP

Properties of Arithmetic Progressions (AP)

1. If a certain number is added or subtracted to each term of a given AP, then the resulting sequence is also an AP and it has the same common difference as that of the given AP

   e.g., 1, 4, 7, 10, 13, 16, …, then

   3, 6, 9, 12, 15, 18, … is also an AP

   and 2, 5, 8, 11, 14, 17, … is also an AP

   In the second sequence each term is increased by 2 than that of the first sequence and each term of the third sequence is decreased by 1 than that of second sequence.

2. If each term of an AP is multiplied by a fixed constant (or divided by a non-zero fixed constant), then the resulting sequence is also an AP *e.g.*, 6, 12, 18, 24, 30, 36, …, then 9, 18, 27, 36, 45, 54, … is also an AP and 3, 6, 9, 12, 15, 18, … is also an AP.

   In the second sequence each term is 3/2 times that of the first sequence and in the third sequence each term is half of the each term of the first sequence.

3. If $m_1, m_2, m_3, \ldots, m_k$ and $n_1, n_2, n_3, \ldots, n_k$ are in AP, then $(m_1 + n_1), (m_2 + n_2), (m_3 + n_3), \ldots, (m_k + n_k)$ are also in AP

   e.g., $S_1 = 3, 7, 11, 15, 19, \ldots$ and $S_2 = 5, 11, 17, 23, 29, \ldots$,

   then $S_1 + S_2 = (3+5), (7+11), (11+17), (15+23), (19+29) \ldots \Rightarrow S_1 + S_2 = 8, 18, 28, 38, 48, \ldots$ is also an AP

4. If $m_1, m_2, m_3, \ldots, m_k$ and $n_1, n_2, n_3, \ldots, n_k$ are in AP, then $(m_1 - n_1), (m_2 - n_2), (m_3 - n_3), \ldots, (m_k - n_k)$ are also in AP *e.g.*, $S_1 = 7, 12, 17, 22, 27, \ldots$, $S_2 = 4, 13, 22, 31, 40, \ldots$. Then $S_1 - S_2 = (7-4), (12-13), (17-22), (22-31), (27-40) \ldots$

   $\Rightarrow$ $S_1 - S_2 = 3, -1, -5, -9, -13, \ldots$ is also an AP

5. If there are odd number of terms in an AP as $a_1, a_2, a_3, \ldots, a_n$, then the middle most term is the A.M. of the given sequence. *e.g.*, 4, 10, 16, 22, 28

   Then, $\frac{4+10+16+22+28}{5} = 16$

   Also $\frac{4+28}{2} = \frac{10+22}{2} = 16\,(\text{AM})$

   (It is the most important point for logical problems.)

6. If there are even number of terms in an AP as $a_1, a_2, a_3, \ldots, a_n$, then

   $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = a_4 + a_{n-3}$ etc.

   e.g., 4, 10, 16, 22, 28, 34, 40, 46

   Then, $4 + 46 = 10 + 40 = 16 + 34 = 22 + 28$

   also $\frac{4+46}{2} = \frac{10+40}{2} = \frac{16+34}{2} = \frac{22+28}{2} = 25\,(\text{A.M.})$

7. The sum of an AP with n terms is given by

   $S_n = n\,(\text{A.M.})$

   e.g. (1) 3, 7, 11, 15, 19, then $S_5 = 5 \times 11 = 55$

   (2) 5, 8, 11, 14, 17, 20

   $\therefore$ $S_6 = 6 \times \frac{25}{2} = 75$

8. If we have to take three terms in AP it is convenient to take them as $(m-k), m, (m+k)$ where $(m-k), m$ and $(m+k)$ are consecutive terms of an AP and k is the common difference. Similarly, we can take five terms in AP as $(m-2k), (m-k), m, (m+k), (m+2k)$.

Exp. 4) If the sum of five terms of an AP is 75, then find the third term of the AP

Solution $(m-2k)+(m-k)+m+(m+k)+(m+2k)=75$

$\Rightarrow \quad 5m=75$

$\Rightarrow \quad m=15$

Hence, the third term (*i.e.* middle most term) is 15.

NOTE In general to take odd number of terms, we assume m as the middlemost term and take other terms with common difference k, as per their position.

9. If we have to take 4 terms in AP, it is convenient to take them as $(m-3k), (m-k), (m+k), (m+3k)$

where $(m-3k), (m-k), (m+k), (m+3k)$ are consecutive terms of an AP and '$2k$' is the common difference. Similarly, we can take six terms in AP as $(m-5k),(m-3k), (m-k), (m+k), (m+3k),(m+5k)$.

Exp. 5) If the sum of 4 terms of an AP is 36, then find the A.M. of the AP

Solution $(m-3k)+(m-k)+(m+k)+(m+3k)=36$

$\Rightarrow \quad 4m=36$

$\therefore \quad S_4=4m=36$

$\therefore \quad \text{A.M.}=\dfrac{S_4}{4}=\dfrac{4m}{4}$

$=\dfrac{36}{4}=9$

10. The sum of first n natural numbers

$$\Sigma n=1+2+3+\ldots+n=\frac{n(n+1)}{2}$$

Exp. 6) Find the sum of first 10 natural numbers.

Solution $\Sigma 10=1+2+3+\ldots+10=\dfrac{10\times 11}{2}=55$

11. The sum of squares of first 'n' natural numbers

$$\Sigma n^2=1^2+2^2+3^2+\ldots+n^2$$

$$=\frac{n(n+1)(2n+1)}{6}$$

Exp. 7) Find the sum of squares of first 11 natural numbers.

Solution $\Sigma n^2=1^2+2^2+3^2+\ldots+11^2=\dfrac{11\times 12\times 23}{6}=506$

12. The sum of cubes of first 'n' natural numbers

$$\Sigma n^3=1^3+2^3+3^3+\ldots+n^3=\left[\frac{n(n+1)}{2}\right]^2=[\Sigma n]^2$$

Exp. 8) Find the sum of cubes of first 10 natural numbers.

Solution $\Sigma(10)^3=1^3+2^3+3^3+\ldots+10^3=\left(\dfrac{10\times 11}{2}\right)^2=3025$

13. The sum of first 'n' odd numbers

i.e., $\quad 1+3+5+7+\ldots+(2n-1)=n^2$

Exp. 9) Find the sum of $1+3+5+7+\ldots+11$

Solution $1+3+5+7+\ldots+11=(11)^2=121$

14. The sum of first n even numbers

i.e., $\quad 2+4+6+\ldots+2n=n(n+1)$

Exp. 10) Find the sum of first 10 even numbers.

Solution $2+4+6+8\ldots 20=10\times 11=110$

Introductory *Exercise 18.1*

1. Find the 41st term of the progression 3, 8, 13, 18, ...:
(a) 102 (b) 203
(c) 304 (d) none of these

2. Find the 25th term of the AP 10, 6, 2, – 2, – 6, – 10, ... :
(a) – 86 (b) 106
(c) 96 (d) none of these

3. What term of the AP 2, 5, 8, ... is 56 ?
(a) 20 (b) 21
(c) 19 (d) 15

4. What is the 10th term of the sequence 2, 4, ... ?
(a) 18 (b) 20
(c) 1024 (d) can't be determined

5. If the 3rd and 7th terms of an AP are 17 and 27 respectively. Find the first term of the AP :
(a) 9 (b) 12
(c) 14 (d) none of these

6. Find the nth term of an AP whose 6th and 8th terms are 12 and 22 respectively :
(a) $7n-30$ (b) n^2-24
(c) $5n-18$ (d) none of these

7. If 7 times the seventh term of an AP is equal to 11 times its eleventh term, the value of eighteenth term of the AP is
(a) 0 (b) – 8 (c) 18 (d) 77

8. If the pth, qth and rth terms of an AP are a, b, c respectively, then the value of
$a(q-r)+b(r-p)+c(p-q)$ is :
(a) 0 (b) 1 (c) abc (d) pqr

9. If the pth term of an AP is q and its q th term is p, then its mth term is :
(a) $\dfrac{pq}{m}$ (b) $pq-m$ (c) $p+q-m$ (d) pqm

10. Find the sum of the AP 11, 13, 15, ..., 99 :
(a) 2475 (b) 2500 (c) 1122 (d) 1580

11. Find the number of terms in the AP 22, 28, 34, ..., 616
(a) 80 (b) 78 (c) 99 (d) 100

12. Find the sum of 222, 224, 226, ..., 888 :
(a) 185370 (b) 195300 (c) 183000 (d) 899000

13. If the second and seventh terms of an AP are 2 and 22 respectively. Find the sum of first 35 terms :
(a) 2310 (b) 3210
(c) 2130 (d) none of these

14. The 12th term of an AP is – 13 and the sum of the first four terms of it is 24. Find the sum of its first ten terms
(a) – 48 (b) – 26 (c) 0 (d) 52

15. The third term of an AP is $\frac{1}{5}$ and the 5th term is $\frac{1}{3}$. Find the sum to 15 terms of the AP :
(a) 1/15 (b) 3/5 (c) 8 (d) 15

16. How many terms of the AP 1, 4, 7, ... are needed to give the sum 925 ?
(a) 20 (b) 22 (c) 24 (d) 25

17. How many terms of the series $20 + 16 + 12 + \ldots$ amounts to 48 ?
(a) 3 (b) 5
(c) 8 (d) both a and c

18. p, q, r, s, t are first five terms of an AP such that $p + r + t = -12$ and $p \cdot q \cdot r = 8$. Find the first term of the above AP :
(a) 3 (b) 2 (c) 4 (d) – 4

19. The sum of all the terms of the AP $7, 10, 13, \ldots$ l is 1242, where l is the last term of the AP Find the value of l.
(a) 67 (b) 79 (c) 85 (d) 102

20. Find the sum of all the integers between 55 and 5555 which are divisible by 7 :
(a) 678 (b) 786
(c) 876 (d) none of these

21. How many terms are there in AP whose first and fifth terms are – 14 and 2 respectively and the sum is 40 :
(a) 12 (b) 10 (c) 16 (d) 8

22. The first and last terms of an AP are – 7 and 233 and the sum of the AP is 9153. Find the number of terms in the AP :
(a) 83 (b) 81 (c) 49 (d) 99

23. The sum of three numbers in AP is 12 and sum of their cubes is 408. Find the product of the numbers :
(a) 17 (b) 35 (c) 28 (d) 36

24. The series of natural numbers is written as follows :

1
2 3 4
5 6 7 8 9
... ...
... ...

Find the sum of the numbers in the nth row :
(a) $n^3 + 1$ (b) $(2n)^2 - 1$
(c) $n^3 + (n-1)^3$ (d) none of these

25. If you save ₹ 1 today, ₹ 2 the next day, ₹ 3 the succeeding day and so on, what will be your total savings in 365 days ?
(a) 66579 (b) 66795
(c) 56795 (d) none of these

26. The ratio of the 7th to the 3rd terms of an AP is 12 : 5, find the ratio of the 13th to 4th term :
(a) 13 : 7 (b) 4 : 3
(c) 10 : 3 (d) none of these

27. Find the sum of the first hundred even natural numbers divisible by 5 :
(a) 50500 (b) 55000
(c) 50050 (d) 50005

28. If m times the mth term of an AP is equal to n times its nth term, find $(m + n)$ th term of the AP :
(a) – 1 (b) 0 (c) mn (d) 4

29. The sum of the first fifteen terms of an AP is 105 and the sum of the next fifteen terms is 780. Find the common difference of AP :
(a) 4 (b) 3 (c) 6 (d) 5

30. If the first term of an AP is 2 and the sum of the first five terms is equal to one-fourth of the sum of the next five terms, then find the 20th term ?
(a) – 114 (b) – 82
(c) – 112 (d) none of these

31. The sum of the first six terms of an AP is 42. The ratio of the 10th term to the 30th term of AP is $\frac{1}{3}$. Find the 40th term of the AP :
(a) – 60 (b) 20 (c) 39 (d) 80

32. The sum of n terms of two arithmetic series are in the ratio of $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms :
(a) 4 : 3 (b) 5 : 4
(c) 7 : 4 (d) none of a, b, c

33. The sum of three numbers in AP is 15 and sum of their squares is 93. Find the greatest number :
(a) 6 (b) 10
(c) 8 (d) 7

34. If the nth term of an AP is $4n - 1$, find the 30th term and the sum of first 30 terms.
(a) 119 and 1830 (b) 69 and 1665
(c) 119 and 1380 (d) none of these

35. The sum of n terms of a series is $3n^2 + 5n$. Find the value of n if nth term is 152 :
(a) 15 (b) 18
(c) 20 (d) 25

36. Find the number of terms of the AP 98, 91, 84, ... must be taken to give a sum of zero :
(a) 14 (b) 15 (c) 31 (d) 29

37. What is the greatest possible sum of the AP 17, 14, 11, ... :
(a) 51 (b) 73 (c) 57 (d) 75

38. What is the least possible sum of the AP – 23, – 19, – 15, ... :
(a) – 3 (b) – 78 (c) – 87 (d) none

39. Find the sum of all odd numbers of four digits which are divisible by 9 :
(a) 2784491 (b) 2478429
(c) 2754000 (d) 2448729

40. If a, b, c be the pth, qth and rth terms of an AP, then $p(b-c)+q(c-a)+r(a-b)$ equals to :
(a) 0 (b) 1 (c) 3 (d) $\frac{abc}{pqr}$

41. If a, b, c be respectively the sum of first p, q, r terms of an AP, then $\frac{a}{p}(q-r)+\frac{b}{q}(r-p)+\frac{c}{r}(p-q)$ equals
(a) pqr
(b) $ap + bq + cr$
(c) 0
(d) $p + q + r$

42. Divide 20 into four parts which are in AP and such that the product of the first and fourth is to the product of the second and third in the ratio 2 : 3. Find the product of the first and fourth term of the AP :
(a) 12 (b) 16
(c) 20 (d) 25

43. The sum of the first p terms of an AP is q and the sum of the first q terms is p. Find the sum of the first $(p + q)$ terms :
(a) pq (b) $p - q$ (c) $-(p + q)$ (d) 0

18.5 Geometric Progression (G.P.)

Geometric Progression

A sequence $a_1, a_2, a_3, \ldots, a_n$ is said to be in Geometric Progression when

$$\frac{a_2}{a_1}=\frac{a_3}{a_2}=\frac{a_4}{a_3}=\ldots r(\text{say})$$

and $a_1, a_2, a_3, \ldots, a_n$ are all non-zero numbers and r is said to be the common ratio of the G.P. *e.g.*,

(i) $2, 4, 8, 16, \ldots \to r=2$
(ii) $1, 3, 9, 27, 81 \ldots \to r=3$
(iii) $-3, 6, -12, 24, -48, 96 \ldots \to r=-2$
(iv) $\frac{1}{2}, \frac{3}{2}, \frac{9}{2}, \frac{27}{2} \ldots \to = r=3$
(v) $k, kr, kr^2, kr^3, kr^4, \ldots \to r=r$
(vi) $64, 16, 4, 1, \frac{1}{4}, \frac{1}{16}, \ldots \to r=\frac{1}{4}$

General term (or *n*th term) *of* a *G.P.*

Let us look at the following $1, 3, 9, 27, 81, 243, 729, 2187, \ldots$

Each next term in the above sequence is becoming thrice (*i.e.* 3 times) which is called as common ratio.

$$1=1$$
$$3=1\times 3$$
$$9=1\times 3\times 3$$
$$27=1\times 3\times 3\times 3$$
$$81=1\times 3\times 3\times 3\times 3$$
$$243=1\times 3\times 3\times 3\times 3\times 3$$
$$729=1\times 3\times 3\times 3\times 3\times 3\times 3$$

and nth term $= a\times r\times r\times r\times \ldots \times (n-1)$ times $T_n = a\cdot r^{(n-1)}$

$\therefore$ The general term or nth term is represented by T_n

Hence, $T_n = a\cdot r^{(n-1)}$

Also $l = a\cdot r^{(n-1)}$

where l is the last term of the G.P.

Exp. 1) Find the 7th term of the sequence $2, 10, 50, 250, \ldots$

Solution The given sequence is a G.P. with common ratio

$$=5 \qquad \left(\because \frac{10}{2}=\frac{50}{10}=5\right)$$

and first term $(a) = 2$

$\therefore$ $T_7 = 2\cdot(5)^6$ $(\because T_n = a\cdot r^{n-1})$

$T_7 = 31250$

Sum of First *n* terms of a G.P.

Let us look at the following : 3, 6, 12, 24, 48, 96, ...

Let the sum of n terms is denoted by S_n, then

$S_1 = 3$
$S_2 = 3 + 6 = 9,\ S_3 = 3 + 6 + 12 = 21$
$S_4 = 3 + 6 + 12 + 24 = 45$
$S_5 = 3 + 6 + 12 + 24 + 48 = 93$

$\therefore$ $S_n = T_1 + T_2 + T_3 + \ldots + T_n$

$\therefore$ $S_n = \frac{a\cdot(r^n-1)}{(r-1)}$ if $r>1$

and $S_n = \frac{a\cdot(1-r^n)}{(1-r)}$ if $r<1$ also, $S_n = \frac{lr-a}{r-1}$

NOTE The above formulae for S_n fails when $r=1$, as the form $\frac{0}{0}$ is undefined. However,

$$S_n = a + a + a + \ldots \text{ to } n \text{ terms} = n\cdot a$$

Exp. 2) Find the sum of the first 6 terms of the G.P.

$$3, \frac{9}{2}, \frac{27}{4}, \frac{81}{8}, \ldots$$

Solution Here $a = 3, r = \frac{3}{2}$ and $n = 6$, then

$$S_6 = \frac{3\cdot\left[\left(\frac{3}{2}\right)^6 - 1\right]}{\left(\frac{3}{2} - 1\right)} = \frac{3\cdot\left(\frac{729}{64} - 1\right)}{\frac{1}{2}} = \frac{3\cdot\left(\frac{665}{64}\right)}{\frac{1}{2}}$$

$$= \frac{3 \times 665 \times 2}{64 \times 1} = 62.34375$$

Exp. 3) Find the sum of first five terms of the G.P. 1, 2, 4, …

Solution $S_5 = \frac{1\cdot(2^5 - 1)}{(2-1)} = 31$

Geometric Mean (G.M.)

If three quantities are in G.P., the middle one is called the geometric mean (G.M.) of the other two quantities.

If a, m, b are in G.P., then m is the G.M. of a and b

and since $\frac{m}{a} = \frac{b}{m}$

$\therefore \quad m^2 = ab \Rightarrow m = \pm\sqrt{ab} = \pm(ab)^{1/2}$

In general, if $a_1, a_2, a_3, \ldots, a_n$ are in G.P., then

$$\text{G.M.} = (a_1 \,.\, a_2 \,.\, a_3 \,.\, \ldots \,.\, a_n)^{1/n}$$

Exp. 4) Find the G.M. of 4, 12, 36, which are in G.P.

Solution G.M. of 4, 12, 36 $= (4 \times 12 \times 36)^{1/3}$

$$= (1728)^{1/3} = (12^3)^{1/3} = 12$$

Exp. 5) Find the G.M. of 3, 6, 12, 24, 48.

Solution G.M. $= (3 \times 6 \times 12 \times 24 \times 48)^{1/5} = (12^5)^{1/5} = 12$

How to Insert n Geometric Means between a and b If a and b are the first and last term respectively between which we have to insert n geometric means $m_1, m_2, m_3, \ldots, m_n$, then

Common ratio, $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

Thus, $m_1 = a\cdot\left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

$$m_2 = a\cdot\left(\frac{b}{a}\right)^{\frac{2}{n+1}}$$

$$m_3 = a\cdot\left(\frac{b}{a}\right)^{\frac{3}{n+1}}$$

… … …

… … …

$$m_n = a\cdot\left(\frac{b}{a}\right)^{\frac{n}{n+1}}$$

Exp. 6) Insert 3 geometric means between 3 and 48.

Solution $r = \left(\frac{48}{3}\right)^{\frac{1}{3+1}} = (2^4)^{\frac{1}{4}} = 2$

$\therefore \quad m_1 = 3\cdot 2 = 6$

$\Rightarrow \quad m_2 = 3\cdot(2)^2 = 12$

$m_3 = 3\cdot(2)^3 = 24$

Hence the new sequence is 3, 6, 12, 24, 48 where all the terms are in G.P.

Properties of Geometric Progression

1. If each term of a G.P. is multiplied or divided by a certain non-zero constant, then the resulting sequence is also a G.P. e.g., 3, 6, 12, 24, 48, 96,… then 9, 18, 36, 72, 144, 288, … is also a G.P. and $\frac{9}{2}$, 9, 18, 36, 72, 144, … is also a G.P.

   In the second sequence each term is multiplied by 3 and each term of the third sequence is divided by 2 (compare with the second sequence)

2. If $a_1, a_2, a_3, \ldots$ and $b_1, b_2, b_3, \ldots$ are two geometric progressions, then the sequence $a_1b_1, a_2b_2, a_3b_3, \ldots$ is also a G.P.

   e.g., 2, 6, 18, 54, 162, … and 1, 2, 4, 8, 16, … then $(2\times1), (6\times2), (18\times4), (54\times8), (162\times16)\ldots$

   i.e. 2, 12, 72, 432, 2592 is also a G.P.

   (Similarly $\frac{2}{1}, \frac{6}{2}, \frac{18}{4}, \frac{54}{8}, \frac{162}{16}, \ldots$

   i.e., $2, 3, \frac{9}{2}, \frac{27}{4}, \frac{81}{8}, \ldots$ is also a G.P.)

3. If there are odd number of terms in a G.P. as $a_1, a_2, a_3, \ldots, a_n$ then the middle most term is the G.M. of the given sequence. *e.g.*,2, 4, 8, 16, 32 then $(2\times4\times8\times16\times32)^{1/5} = (2^{15})^{1/5} = 2^3 = 8$

   Also, $(2\times32)^{1/2} = (4\times16)^{1/2} = 8$

NOTE In this case without knowing all the terms we can find the G.M. of the G.P.

4. If there are even number of terms in a G.P. as $a_1, a_2, a_3, \ldots, a_n$, then

   $\sqrt{a_1\cdot a_n} = \sqrt{a_2\cdot a_{n-2}} = \sqrt{a_3\cdot a_{n-3}} = \ldots$ G.M.

5. If we have to take three terms in G.P., it is convenient to take them as $\frac{m}{k}, m, mk$; where $\frac{m}{k}$, m and mk are consecutive terms of G.P. and k is the common ratio. Similarly we can take five terms of G.P. as $\frac{m}{k^2}, \frac{m}{k}, m, mk, mk^2$.

Exp. 7) If the product of five consecutive terms of G.P. is 1024, then find the third term of this set.

Solution $\frac{m}{k^2}\times\frac{m}{k}\times m\times mk\times mk^2=1024$

$\Rightarrow \quad m^5=1024=2^{10}=4^5\Rightarrow m=4$

Hence the third term (*i.e.* middle most term) is 4.

In general to take odd number of terms, we assume m as the middle most term and take other terms with common ratio k, as per their position.

6. If we have to take 4 terms in G.P. it is convenient to take them as $\frac{m}{k^3}, \frac{m}{k}, mk, mk^3$ where $\frac{m}{k^3}, \frac{m}{k}, mk$ and mk^3 are four consecutive terms of a G.P. and k^2 is the common ratio. Similarly we can take six terms in G.P. as $\frac{m}{k^5}, \frac{m}{k^3}, \frac{m}{k}, mk, mk^3, mk^5$

Exp. 8) If the product of 4 terms of a G.P. is 729, then find the G.M. of the G.P.

Solution $\frac{m}{k^3}\times\frac{m}{k}\times mk\times mk^3=729$

$\Rightarrow \quad m^4=729=3^6\Rightarrow m=3^{6/4}=3^{3/2}\Rightarrow m=\sqrt{27}$

7. If $a_1, a_2, a_3, \ldots, a_n$ are in G.P. then $a_1^m, a_2^m, a_3^m, \ldots, a_n^m$ are also in G.P.

   e.g. 1, 2, 4, 8, 16, … then 1, 4, 16, 64, 256, … are also in G.P.

8. If $a_1, a_2, a_3, \ldots, a_n$ are in G.P. (for every $a_i>0$; $i\in I^+$ then $\log a_1, \log a_2, \log a_3, \ldots$ are in AP

   (The converse of the above is also true)

Sum of an Infinite G.P.

$$S_\infty=\frac{a}{1-r};\ |r|<1$$

where a is the first term and r is the common ratio of the G.P. of infinite terms.

Exp. 9) Find the sum of an infinite G.P. whose first term is 28 and the second term is 4.

Solution $a=28,\ T_2=ar=4\Rightarrow r=\frac{1}{7}$

$\therefore \qquad S_\infty=\frac{a}{1-r}=\frac{28}{1-\frac{1}{7}}=\frac{28}{6}\times 7=\frac{98}{3}$

Introductory *Exercise 18.2*

1. Find the 22nd term of the G.P. – 2, 2, –2, ...:
(a) – 22 (b) – 2
(c) 2 (d) none of these

2. Find the 7th term of the series $-\frac{1}{8}+\frac{1}{4}-\frac{1}{2}+1\ldots$:
(a) 8 (b) – 16
(c) – 8 (d) none of these

3. The 5th, 8th and 11th terms of a G.P. are a, b, c respectively, then which one of the following is true ?
(a) $2b = ac$ (b) $b^2 = ac$
(c) $a + b + c = 0$ (d) none of these

4. The 5th and 12th terms of a G.P. are 32 and 4096 respectively. Find the nth term of the G.P. :
(a) 2^n (b) n^2
(c) $2n^2$ (d) none of these

5. What is the least number of terms of the G.P. $5 + 10 + 20 + \ldots$ whose sum would surely exceed 10^6?
(a) 17 (b) 18
(c) 19 (d) 21

6. The A.M. of two positive numbers is 15 and their G.M. is 12. What is the smaller number ?
(a) 8 (b) 12
(c) 6 (d) 24

7. The sum of 3 numbers in G.P. is 38 and their product is 1728. Find the greatest number :
(a) 24 (b) 18 (c) 16 (d) 8

8. The sum of first three terms of a G.P. is to the sum of the first six terms is 125 : 152. Find the common ratio of the G.P. :
(a) 3/5 (b) 3
(c) 2/5 (d) 5/8

9. The sum of three numbers in G.P. is 14. If the first two terms are each increased by 1 and the third term is decreased by 1, the resulting numbers are in AP Find the product of these three numbers :
(a) 125 (b) 64
(c) 216 (d) 124

10. The third term of a G.P. is 4. The product of first five terms is :
(a) 4^3 (b) 4^4
(c) 4^5 (d) none of these

11. The sum of first three terms of a G.P. is 21 and sum of their squares is 189. Find the common ratio :
(a) $\frac{1}{2}$ or 2 (b) 3 or $\frac{1}{3}$
(c) 4 or $\frac{1}{4}$ (d) none of these

12. The sum of the first and the third terms of a G.P. is 15 and that of the 5th and the 7th terms is 240. Find the 9th term
(a) 678 (b) 786
(c) 867 (d) 768

13. Sum of three consecutive terms in a G.P. is 42 and their product is 512. Find the largest of these numbers
(a) 28 (b) 16
(c) 32 (d) none of these

14. The sum of three numbers in G.P. is 70, if the two extremes be multiplied each by 4 and the mean by 5, the new numbers so formed are in AP Find the product of original numbers :
(a) 8000
(b) 6000
(c) 7000
(d) none of the above

15. The sum of four terms in G.P. is 312. The sum of first and fourth terms is 252. Find the product of second and third terms :
(a) 500 (b) 150
(c) 60 (d) none of these

16. A bouncing tennis ball rebounds each time to a height equal to one half the height of the previous bounce. If it is dropped from a height of 16 m, find the total distance it has travelled when it hits the ground for the 10th time :
(a) $47\frac{15}{16}$ (b) $37\frac{5}{16}$
(c) $67\frac{11}{16}$ (d) none of these

17. Find the sum to n terms of the series
$$7 + 77 + 777 + 7777 + \ldots$$
(a) $\frac{7}{9}\{10(10^n - 1) - n\}$
(b) $\frac{7}{9}\left\{\frac{10}{9}(10^n - 1) - n\right\}$
(c) $7\left\{\frac{10}{9}(10^n - 1) - n\right\}$
(d) none of the above

18. Find the sum to n terms of $0.8 + 0.88 + 0.888 + \ldots$
(a) $\frac{8}{9}\left[\frac{1}{9}\{1 - (0.1)^n\}\right]$ (b) $\frac{8}{9}\left[\frac{1}{10}\{1 - (0.1)^n\}\right]$
(c) $\frac{8}{9}\left[n - \frac{1}{9}\{1 - (0.1)^n\}\right]$ (d) none of these

19. If the pth, qth and rth terms of a G.P. be respectively a, b, and c then the value of $a^{q-r} \cdot b^{r-p} \cdot c^{p-q}$ is :
(a) 0 (b) 1
(c) 3 (d) $\frac{abc}{pqr}$

20. If a, b, c are respectively the xth, yth and zth terms of a G.P, then the value of
$(y - z)\log a + (z - x)\log b + (x - y)\log c$:
(a) 0 (b) 1
(c) 3 (d) none of these

21. There are four numbers such that the first three of them form an Arithmetic Progression and the last three form a Geometric Progression. The sum of the first and the third is 2 and that of second and fourth is 26. What is the sum of first and fourth numbers?
(a) $85\frac{13}{25}$ (b) $25\frac{73}{85}$
(c) 83 (d) none of these

22. Find the sum to n terms of the series
$1 + 3 + 7 + 15 + \ldots$
(a) $[(2^n - 1) - n]$ (b) $[(2^n - 1) + n]$
(c) $[2(2^n - 1) - n]$ (d) none of these

23. Find the sum of $\frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \ldots + \frac{2n-1}{2^n}$:
(a) $\left(\frac{2n-3}{2^n}\right)$ (b) $\left(3 - \frac{2n+3}{2^n}\right)$
(c) $\left(\frac{2n+3}{2^n}\right)$ (d) none of these

24. Find the sum to n terms of the series
$$11 + 102 + 1003 + 10004 + \ldots :$$
(a) $(10^n - 1) + \frac{n(n+1)}{2}$
(b) $\frac{10}{9}(10^n - 1) + \frac{n(n+1)}{2}$
(c) $10^n + n^2 - 1$
(d) none of the above

25. Find the sum to first n groups of
$(1) + (1 + 3) + (1 + 3 + 9) + (1 + 3 + 9 + 27) + \ldots$
(a) $\frac{1}{2}(3^n - 1)$ (b) $\frac{3}{4}(3^n - 1) - \frac{n}{2}$
(c) $n^2 + 1$ (d) none of these

26. Find the sum to n terms of the following series :
$$2 + 5 + 14 + 41 + \ldots$$
(a) $\frac{3}{4}(3^{n-1}) + \frac{n}{2}$ (b) $\frac{3}{4}(3^n - 1) + \frac{n}{2}$
(c) $(3n + 1)$ (d) none of these

27. Find the sum to n terms :
$$1 + 2x + 3x^2 + 4x^3 + \ldots; x \neq 1 :$$
(a) $1 + nx^n$ (b) $\frac{x(x+1)}{2} + x^n$
(c) $\frac{1 - x^n}{(1 - x)^2}$ (d) $\left\{\frac{1 - x^n}{(1 - x)^2} - \frac{nx^n}{(1 - x)}\right\}$

28. Find the sum to n terms of $1 + 3x + 5x^2 + 7x^3 + \ldots$, $x \neq 1$.

(a) $2(1 - x^{n-1})$ (b) $\dfrac{2x(1 - x^{n-1})}{(1 - x^2)}$

(c) $\dfrac{2x(1 - x^{n-1})}{(1 - x)^2} + \dfrac{1 - (2n - 1)x^n}{(1 - x)}$

(d) none of these

29. Find the sum to first n terms :

$$1 + \frac{2}{3} + \frac{3}{3^2} + \frac{4}{3^3} + \ldots$$

(a) $\dfrac{3}{4} - \dfrac{3 + 2n}{4(3^{n-1})}$ (b) $\dfrac{9}{4} - \dfrac{3}{4}\left[\dfrac{3 + 2n}{3^n}\right]$

(c) $2n - \left(\dfrac{1}{n}\right)^2$ (d) none of these

30. Find the sum to n terms of

$$3 \cdot 2 + 5 \cdot 2^2 + 7 \cdot 2^3 + \ldots$$

(a) $2^{n+2} - 2^{n+1} - 2$ (b) $n \cdot 2^{n+2} - 2^{n+1} + 2$

(c) $n \cdot 2^{n+2} + 2^{n-2}$ (d) none of these

31. Find the sum of the series :

$$1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \ldots \text{ to } n \text{ terms :}$$

(a) $\dfrac{35}{16} + \dfrac{7n}{5}$ (b) $\dfrac{12n - 7}{16(5^{n-1})}$

(c) $\dfrac{35}{16} - \left[\dfrac{12n + 7}{16(5^{n-1})}\right]$ (d) none of these

32. Find the sum of the series

$$1 \cdot 3^2 + 2 \cdot 5^2 + 3 \cdot 7^2 + \ldots \text{ to 20 terms :}$$

(a) 12896 (b) 187898

(c) 98970 (d) 188090

33. Find the sum of the series

$$\frac{1^3}{1} + \frac{1^3 + 2^3}{1 + 3} + \frac{1^3 + 2^3 + 3^3}{1 + 3 + 5} + \ldots \text{ to 16 terms}$$

(a) 224 (b) 446

(c) 2356 (d) none of these

34. In a set of four numbers, the first three are in G.P. and the last three are in AP with common difference 6. If the first number is the same as the fourth, find the third number :

(a) – 8 (b) 4

(c) 2 (d) can't be determined

35. The sum of an infinite G.P. is 16 and the sum of the squares of its terms is $153\dfrac{3}{5}$. Find the fourth term of the progression :

(a) $\dfrac{7}{16}$ (b) $\dfrac{5}{32}$

(c) $\dfrac{3}{16}$ (d) none of these

36. If $x = 1 + a + a^2 + \ldots$ to ∞ and $y = 1 + b + b^2 + \ldots$ to ∞, and $|a| < 1, |b| < 1$, then the value of :

$1 + ab + a^2b^2 + \ldots$ to ∞ is :

(a) $\dfrac{xy}{x + y}$ (b) $\dfrac{x + y + 1}{xy}$

(c) $\dfrac{xy}{x - y}$ (d) $\dfrac{xy}{x + y - 1}$

37. A person is entitled to receive an annual payment which for each year is less by one tenth of what it was for the year before. If the first payment is 100, then find the maximum possible payment which he can receive, however long he may live :

(a) 900

(b) 9999

(c) 1000

(d) none of the above

38. Find the sum of the series $\dfrac{1}{4} - \dfrac{3}{16} + \dfrac{9}{64} - \dfrac{27}{256} + \ldots$

(a) $\dfrac{1}{3}$ (b) $\dfrac{1}{4}$

(c) $\dfrac{1}{7}$ (d) none of these

39. The sum of first two terms of a G.P. is $\dfrac{5}{3}$ and the sum to infinity of the series is 3. Find the first term :

(a) 1 (b) $\dfrac{2}{3}$

(c) 5 (d) both (a) and (c)

40. A ball is dropped from a height of 96 ft and it rebounds $\dfrac{2}{3}$ of the height it falls. If it continues to fall and rebound, find the total distance that the ball can travel before coming to rest.

(a) 240 ft (b) 360 ft

(c) 480 ft (d) none of these